

Nodes : π (destructive), Antinodes : 0 (constructive)

- 18 A coil contains N turns of insulated copper wire wound on to a cylindrical iron core of diameter D . The copper wire has a diameter d . The resistivity of copper is ρ . Diameter D is much greater than diameter d .

What is the total resistance between the two ends of the coil?

A $\frac{4N\rho D}{d^2}$ B $\frac{4N\rho d}{D^2}$ C $\frac{8N\rho D}{d^2}$ D $\frac{8N\rho d}{D^2}$

Ans: A

$$R = \rho \frac{L}{A} = \rho \frac{N(\pi D)}{\pi(\frac{d}{2})^2} = \frac{4N\rho D}{d^2}$$

- 19 The circuit diagram shows four resistors of different resistances P , Q , R and S connected to a